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Are shadow banks a threat to the financial stability of EMEs?

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Abstract

Purpose – Shadow banks or non-bank financial intermediaries (NBFIs) are facilitators of credit, especially in emerging market economies (EMEs). However, there are certain risks associated with them, such as their unchecked leverage and interconnectedness with the rest of the financial system. In light of this, the present study analyses the impact of the growth of shadow banks on the stability of the banking sector and the overall stability of the financial system. The authors further examine the effect of the growth of finance companies (a type of NBFIs) on financial stability.

Design/methodology/approach – The study employs data of 11 EMEs (monitored by the Financial Stability Board (FSB)) for the period 2002–2020 to examine the above relationships. Panel-corrected standard errors method and Driscoll–Kray standard error estimation are deployed to conduct the analysis.

Findings – The results signify that the growth of the shadow banking sector and the growth of lending to the shadow banking sector are negatively associated with the stability of the banking sector and increases the vulnerability of the financial system (overall instability). This implies that the higher the growth of the shadow banks, the higher the financial fragility. Finance companies are also found to negatively affect financial stability. These findings are validated by different estimation methods and point out the risks posed by the NBFI sector.

Originality/value – The extant study builds a composite index (Financial Vulnerability Index (FVI)) to measure financial stability; thus, the findings contribute to the evolving literature on shadow banks.

Keywords Shadow banks, NBFIs, Financial stability, PCSE, FSB

Paper type Research paper

1. Introduction

Non-bank financial intermediaries (NBFIs) are engaged in credit intermediation, but unlike banks, they are not covered by explicit guarantees from the government. NBFIs are considered as an alternative source of finance to banks, and thus, they promote the economic growth of a nation, especially in emerging market economies (EMEs).

The global financial crisis (GFC) of 2008 underscored certain risks posed by the NBFI sector. Pro-cyclicality (Adrian & Ashcraft, 2012; Huang, 2018), liquidity mismatches, dependence on leverage for their funding (Barth, Li, Shi, & Xu, 2015; Anshuman & Sharma, 2020), financial layering of loans (Bouguelli, 2020) and asset price fluctuations (Bengtsson, 2013) are some risks that NBFIs are exposed to. Due to their interlinkages, these risks spilled to the other parts of the financial system. NBFIs were thus termed shadow banks. The term "shadow banks" was coined by the economist Paul McCulley (Duca, 2014). NBFIs were called so since they are engaged in bank-like activities. However, they are different from the latter as they are not regulated like banks.

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In order to ensure financial discipline, the regulators set up the Financial Stability Board (FSB) in the aftermath of the GFC. "FSB has been administering NBFIs since 2011, covering 29 economies (constituting 80% of the global gross domestic product (GDP)), including advanced and developing nations" (Arora & Kashiramka, 2023). The growth of NBFIs in EMEs outweighed the growth of NBFIs in advanced countries (FSB, 2020). Despite several regulations in place, shadow banks continue to grow and their unchecked growth could be detrimental to the financial stability of the nation as the risks attached to the NBFI sector may spill over to the other sectors in the financial system owing to their interlinkage with banks and other financial institutions (OFIs).

The present study, thus, analyses the effect of the growth of the NBFI sector on the financial stability of 11 EMEs monitored by the FSB for the period 2002–2020. The study further assesses this relationship for finance companies (a type of NBFIs) for the sampled EMEs. The study makes the following unique contributions to the literature. Firstly, the current literature relating to the effect of the growth of shadow banks on financial stability is scarce, especially in EMEs. Secondly, finance companies are an important type of NBFIs in emerging nations such as India, Indonesia, Russia and Turkey (FSB, 2020). These NBFIs rely on short-term wholesale funding from banks and OFIs. Thus, they may pose stability risks to the financial system due to their activities. Thirdly, the study measures financial stability from the viewpoint of banking sector stability as well as developing an overall measure to gauge financial instability. This measure is a composite measure of 12 indicators representing four sectors: the banking sector, non-financial sector, external sector and valuation pressures.

The findings indicate that the growth of shadow banks causes financial vulnerability, corroborating the findings in the extant literature (Bengtsson, 2013; Chaturvedi & Singh, 2022). Furthermore, the growth of finance companies is also found to be causing financial instability in EMEs. This finding implies that the higher the growth of NBFIs, the higher the financial fragility.

The remaining paper is structured as follows: section 2 explains the concept of NBFIs, section 3 reviews the literature, section 4 discusses the data and explains the methodology, section 5 presents the results of the study, section 6 explains the robustness analysis, section 7 discusses the overall findings and concludes.

2. Concept of NBFIs

This section discusses the concept and relevance of NBFIs in EMEs. There is no definite definition for NBFIs as they vary with respect to the type and complexity of activities across nations (IMF, 2014). However, regulatory bodies and literature have attempted to define such institutions in several ways. The FSB defines NBFIs (broadly) as "the system of credit intermediation that involves entities and activities outside the regular banking system" (FSB, 2011). NBFIs, thus, include all the entities providing credit except banks. Specifically, they are classified by the FSB into pension funds, insurance companies and OFIs. OFIs are further classified as hedge funds, finance company, investment funds and broker-dealers.

Literature holds these institutions amongst others as the culprits for the occurrence of the GFC of 2008 due to their activities and limited regulation (Pozsar, Adrian, Ashcraft, & Boesky, 2010/2012; Lysandrou & Nesvetailova, 2015). Shadow banks are also defined as institutions that are engaged in credit intermediation activities that are "implicitly enhanced, indirectly enhanced or unenhanced by official guarantees established on an *ex ante* basis" (Adrian & Ashcraft, 2012).

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2.1 NBFIs in the emerging nations

The FSB gauges the performance of NBFIs on an annual basis (including pension funds and insurance companies) in 29 advanced and emerging markets. NBFIs in EMEs constituted 72% share of their GDP and 11% of the global NBFI assets in 2020 (FSB, 2021). The relevance of these entities has increased as they have evidenced faster growth than advanced nations, especially during 2013–2020, albeit slower growth than banks and central banks in the year 2020 due to the coronavirus pandemic.

NBFIs in EMEs are less complex than those in advanced nations, and they mainly include finance companies, investment funds, credit unions and insurance firms (Ghosh, Mazo, & Robe, 2012). The growth of NBFIs in 2020 outweighs their compounded annual growth in 2014–2019 in Argentina, Brazil, Russia, Saudi Arabia and Turkey. However, there was a decline and even negative growth in Chile (−3.4%) and Indonesia (−7.03%), which may have been caused by the onset of the pandemic.

The OFIs represented approximately 20% of total financial assets (TFA) in these EMEs [1]. They also represented a major type of NBFIs in these economies. Furthermore, OFIs represented a significant share as a percentage of GDP, particularly in nations such as Brazil (137%), China (88%) and India (36%). OFIs, thus, are vital entities in the financial system of EMEs. However, post the GFC of 2008, there was a realization of the financial stability risks these entities may expose to a nation. These entities are prone to the risk of pro-cyclicality and dependence on short-term funding (Adrian & Ashcraft, 2012).

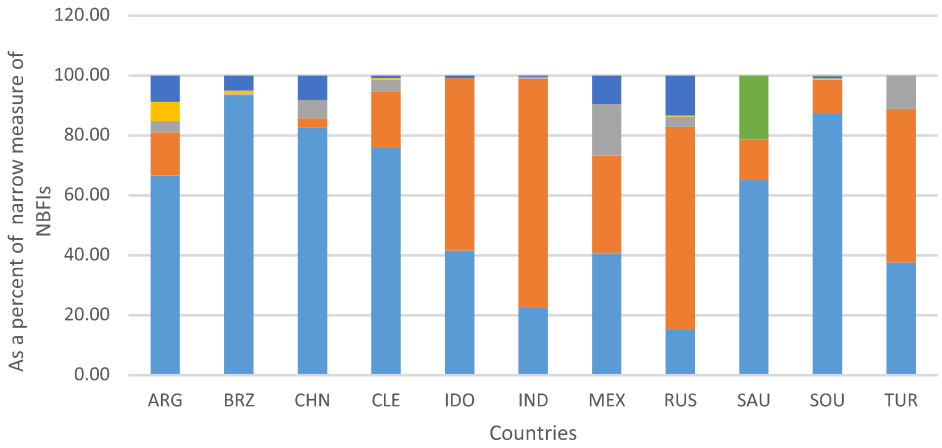
Furthermore, OFIs in EMEs are interlinked with other financial entities in the financial system, which may enhance the interconnectedness risks of these economies. For instance, OFI borrowings from banks exceeded 10% of OFI financial assets in the cases of Russia (21%),Indonesia (19%), Turkey (13%) and India (12%) in 2020. OFI exposure to banks (as a percentage of domestic OFI assets) exceeded 20% in the cases of Argentina (36%), South Africa (42%) and Chile (22%). They are also dependent on pension funds, insurance companies and OFIs for their funding (FSB, 2021).

2.2 Finance companies

In order to capture the vulnerabilities NBFIs may pose due to their excessive leverage or maturity transformation, the FSB narrows down this broad measure to study the activities that may pose risks to the financial system. This narrow measure is classified into the following economic functions (EFs). EF1 includes collective investment vehicles that are likely to face risk related to runs. EF2 refers to NBFIs that rely on short-term funding for their lending. This function includes factoring companies, finance companies and consumer credit NBFIs. EF3 includes NBFIs involved in market intermediation who rely on short-term funding. EF4 refers to NBFIs that provide credit creation, and EF5 refers to NBFIs that undertake securitization activities.

Figure 1 illustrates that a significant portion of the overall narrow measure of NBFIs is either EF1 or EF2. EF1 is majorly represented by money market funds (MMFs) and fixed-income funds. These instruments diversify the risks of investors and allow them to invest in a portfolio of assets. However, some of these entities may assume maturity transformation or depend on excessive leverage, which makes them susceptible to runs. The FSB states that EF2 assets are growing in relevance in EMEs, especially in India as they represent more than one-fourth of the global EF2 assets in 2019 as against 10% in 2013. They comprised a significant portion of the narrow measure in EMEs such as India (77%), Russia (68%), Indonesia (58%) and Turkey (51%) in 2020, while it was of secondary importance for economies such as Mexico (33%), Chile (19%), Argentina (14%), Saudi Arabia (13%) and South Africa (11%). Furthermore, finance companies are the key entity of EF2 as they constitute approximately 80% of the global EF2 assets.

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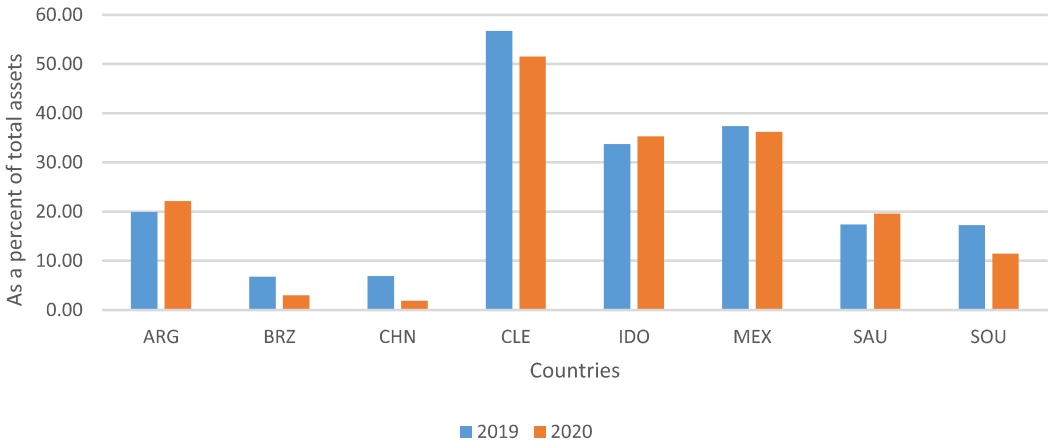
Note(s): Country labels: ARG-Argentina, BRZ-Brazil, CHN-China, CLE-Chile, IDO- Indonesia, IND- India, MEX- Mexico, RUS- Russia, SAU- Saudi Arabia, SOU- South Africa and TUR- Turkey

Source(s): Global Monitoring Report on NBFI 2021, FSB (FSB, 2021)

Figure 1.
Classification of economic functions (EFs) of the FSB as at the end of 2020

Figure 2 shows the dependence of finance companies on short-term wholesale funding in EMEs. Moreover, finance companies in EMEs such as Brazil, Chile and Mexico rely on banks for their funding [2], while Indian finance companies are dependent on OFIs for their funding, thereby raising signs of interconnectedness.

Thus, the interlinkage between NBFIs and other entities in the financial system highlights the risks of financial instability. Moreover, the relevance of finance companies and EF2 assets in EMEs has increased over the years. Thus, in light of the risks attached to the NBFI sector, the present study assesses the relationship between NBFIs and financial stability for EMEs.



Note(s): Country labels: ARG-Argentina, BRZ-Brazil, CHN-China, CLE-Chile, IDO- Indonesia, MEX- Mexico, SAU- Saudi Arabia and SOU- South Africa

Source(s): Global Monitoring Report on NBFI 2021, FSB (FSB, 2021)

Figure 2.
Short-term wholesale funding of finance companies

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3. Literature review and hypothesis development

3.1 Theoretical background

Financial stability is defined as "the capability of a financial system to absorb shocks without affecting the financial markets and institutions" (Motelle & Biekpe, 2015). Allen and Wood (2006) define it as a situation that does not cause serious macroeconomic effects.

NBFIs contribute to economic growth (Barth *et al.*, 2015) and financial inclusion (Knaack & Gruin, 2021) and are considered as an alternative source of finance (Ilesanmi & Tewari, 2019) in EMEs. However, there are several risks that the NBFI sector may be exposed to, which may cause financial instability. Several researchers post the GFC of 2008 discussed the relationship between shadow banking and financial stability.

Bengtsson (2013) explained the mechanisms through which MMFs may create financial instability for the other parts of the financial system. The GFC of 2008 highlighted the instability inherent in MMFs, which may spill over to the banking sector and thereby the entire financial system. Barth *et al* (2015) describe the components of shadow banking in China and their effect on economic growth. They posit that NBFIs are contributors to economic growth. However, there are systemic vulnerabilities and risks attached to their activities. There are interlinkages between banks and non-banks. This may cause banks to suffer when the latter faces problems and thus cause systemic risks. Ilesanmi & Tewari (2019) discussed the alternative role of shadow banks in credit intermediation in South Africa. They pointed out, however, the risks attached to the sector such as liquidity mismatch. Huang (2018) modelled shadow banking activities as the off-balance sheet financing activities of banks and highlighted the pro-cyclicality of the shadow banking sector, which may lead to financial instability. Bouguelli (2020) emphasized the financial fragility caused by shadow banks. Shadow banks such as special purpose vehicles (SPVs) create a financial layering of the loans in the financial system, and this layering further leads to the interdependence of financial institutions. Thus, the unstable nature of the NBFI sector may hamper financial stability.

3.2 Empirical research

Post the GFC of 2008, financial stability has received much attention. The literature on financial stability encompasses the banking sector, macro-economic aspects, developing measures of financial stability and lately the shadow banking sector.

3.2.1 Banking sector and financial stability. Since banks are an important credit intermediary in a financial system, banking stability has long been the focus of many studies. Factors such as bank concentration and increased market power (Uhde & Heimeshoff, 2009; Ozili, 2018), financial inclusion (Cihak, Mare, & Melecky, 2021) and differences between foreign and domestic banks with respect to stability (Haas & Lelyveld, 2006) have been analysed in the banking literature.

Several researchers analysed the link between banking concentration and financial stability. Increased concentration will lead to few large banks, which are believed to promote diversification, better credit monitoring, increased profits and economies of scale and will thus ensure financial stability ("concentration-stability"). On the contrary, market concentration may create few large banks and thus create monopoly powers, complexity, higher risks and thus lower financial stability ("concentration-fragility") (Uhde & Heimeshoff, 2009). Berger, Klapper, and Ariss (2009) assessed the link between competition and financial stability for banks from 23 developed nations. They found that banks with higher concentrations have lower overall risk. Uhde and Heimeshoff (2009) studied the effect of bank concentration on the financial stability of the European Union. They find that the concentration reduces financial soundness. Also, lower diversification opportunities and government ownership tend to increase instability, while capital regulation helps to reduce it. Soedarmono, Machrouh, and

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Tarazi (2011) also analyse the effect of banking consolidation (and thereby increasing banking market power) on financial stability in 12 Asian markets including India. They employ the insolvency risk, income volatility and capitalization of banks as a measure of financial stability. They found bank market power and financial stability (measured as insolvency risk and income volatility) to be negatively related. However, they also report that the higher market power of banks ensured higher capitalization and thus greater financial stability. Boulanouar, Alqahtani, and Hamdi (2021) highlight the importance of the stability of the banking sector for overall financial stability.

3.2.2 Macro-economic aspects and financial stability. The literature examining the macro-economic aspects and financial stability has focussed on various perspectives such as the relationship between financial stability and financial integration (Motelle & Biekpe, 2015), inflation targeting and financial stability (Woodford, 2012), foreign direct investment (FDI) trends and financial stability risks (Kellard *et al.*, 2022) and monetary policy and financial stability (Tobal & Menna, 2020; Eslamloueyan & Fatemifar, 2021).

Motelle and Biekpe (2015) analyse the link between financial integration and financial instability in the South African Development Community consisting of 10 nations for 26 years. They measure the financial instability by modelling the volatility of macro-economic variables, namely output, real exchange rates, inflation, credit and real interest rates. They found that financial integration causes volatility in credit, real interest rates and output, while inflation volatility led to financial integration. Sui, Liu, Li, and Zhang (2022) employ stock market volatility as a measure of financial stability in China. They found that monetary and macroprudential policies together ensure financial stability. Kellard *et al.* (2022) measured the effect of financial system risks on FDI trends for 112 countries. They proxy financial stability as the banking sector and sovereign risks. Banking sector risks measured in terms of the non-performing assets (NPA) ratio affect the FDI flows in origin countries, while sovereign risks affect the FDI flows in both host and origin countries. They highlight the impact of crisis periods such as the GFC and sovereign crisis on the capital flows.

3.2.3 Measures of financial stability. Financial stability has been measured using different proxies in the literature such as bank Z score (Soedarmono, Machrouh, & Tarazi, 2011; Diallo & Mansour, 2017), bank NPA ratio (Ozili, 2018; Kellard *et al.*, 2022) and stock market volatility (Akter & Nobi, 2018; Sui *et al.*, 2022). In order to measure financial stability, several authors such as (Hawkins & Klau, 2000; End, 2006; Gadanecz & Jayaram, 2009; Roy, Biswas, & Sinha, 2015; Aikman, Kiley, Lee, Palumbo, & Warusawitharana, 2017; Lee, Posenau, & Stebunovs, 2017, Lee, Posenau, & Stebunovs, 2020; Akosah, Loloh, Lawson, & Kumah, 2018; Lepers & Serrano, 2020) have developed an index to measure financial stability.

Aikman *et al.* (2017) point out the inability of a single measure to identify the risks emanating in a financial system. Thus, they deployed 46 indicators to measure financial vulnerabilities covering valuation pressures, financial sector and non-financial sector in the US. They found that the Financial Vulnerability Index (FVI) granger causes the credit to GDP gap, which is a widely used measure of financial vulnerability. Similarly, Lee, Posenau, and Stebunovs (2020) studied the ability of the FVI to predict the banking crisis. They developed an FVI for 27 advanced and emerging nations. The index is based on five sub-indices of vulnerability: external sector, sovereign vulnerability, non-financial sector, financial sector and risk appetite. They purported that their aggregate index has better predictive power of a banking crisis than sector-specific indices. Lepers and Serrano (2020) developed financial instability measure or FVI for 11 emerging nations. They deployed 32 indicators in order to holistically measure financial stability. They broadly covered the dimensions of the financial sector, external sector, valuation and risk appetite, and non-financial sector imbalances. They pointed out the deepening of financial systems of emerging nations and the need to study financial stability from domestic as well as global perspectives for these nations.

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3.2.4 Shadow banking and financial stability. There are limited studies on the effect of the NBFI sector on financial soundness. Most of the extant empirical literature has focused on either a single activity (Aviles, Pineira, Caceres, Vergara, & Araya, 2021) or a single economy (Chaturvedi & Singh, 2022) to study whether shadow banking is detrimental to financial stability.

Diallo and Mansour (2017) examined the link between the insurance sector and the NBFI sector and financial stability for 26 nations. They found the insurance sector to be more adverse to financial stability for nations with high levels of shadow banking. Aviles *et al.* (2021) studied the impact of liquidity mismatches in mutual funds on the systemic risks of the financial system in Chile. They found that the effect of liquidity mismatches of mutual funds on the systemic risks was not influential. Anshuman and Sharma (2020) analysed the dependence of retail NBFCs in India on liquid debt mutual funds. They found the former to be dependent on short-term funding which led to the transfer of risks from the former to the latter. They also constructed a health score to predict the financial fragility of these NBFCs. Chaturvedi and Singh (2022) studied the interconnectedness and systemic risks of the NBFI sector in India. They deployed Granger causality and based their research on 2018–19 crisis period of Indian NBFCs (failure of a large NBFC led to crisis in Indian financial markets). They found that there was high interconnectedness during the crisis period among these entities. Moreover, larger, more connected and prestigious institutions suffered fallouts in their market capitalizations.

As pointed out by Bengtsson (2013), the financial stability literature mainly surrounds the banking sector. Although, of late, research in the shadow banking sector is evolving, there is a dearth of empirical studies focussing on the link between shadow banking and financial stability. Thus, it is vital to study the effect of the NBFI sector on the overall financial stability. The present study attempts to investigate the relationship between the shadow banking sector and financial stability. Furthermore, the study narrows down to gauge an important and rapidly growing component of the shadow banking system (especially in EMEs), EF2. The EF2 function of FSB mainly comprises finance companies, and the study attempts to examine the effect of the growth of these finance companies on the financial stability of EMEs.

3.3 Hypothesis development

Post the GFC of 2008, various studies highlighted the detrimental effect of shadow banking on the financial stability of economies due to their asset–liability mismatches and lack of explicit government guarantees (Adrian & Ashcraft, 2012; Bengtsson, 2013). Empirical evidence shows them to be dependent on short-term funding and highly interconnected with OFIs (Anshuman & Sharma, 2020; Chaturvedi & Singh, 2022; Verma & Chakarwarty, 2023).

However, another strand of literature reported that NBFIs complement banks and provide financial services to the unbanked population in emerging economies such as India and China (Arora & Zhang, 2018). Aviles *et al.* (2021) did not find liquidity mismatch of mutual funds to affect the systemic risks in Chile. Thus, the hypothesis is stated as follows:

H1. Growth of NBFIs has a significant relationship with financial stability.

Literature has evidenced a positive relationship between bank profitability and financial stability (Pawlowska, 2016; Ozili, 2018). The profitable banks are expected to be more stable. Thus, the hypothesis is stated as follows:

H2. Bank profitability has a significant positive relationship with financial stability.

Literature indicates that the link between bank concentration and financial stability can be either positive or negative due to the "concentration-stability" (positive) and "concentration-fragility" views (negative). Highly concentrated banking systems and low competition promote economies of scale, risk diversification and yield higher profits. Moreover, a few

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large banks are also easy to monitor, which thus ensures financial stability ("concentration-stability" view). However, large banks give rise to complex organizations and risky investments ("too-big-to-fail" crisis), which may have an adverse impact on financial stability ("concentration-fragility" view) (Berger, Klapper, & Ariss, 2009; Uhde and Heimeshoff, 2009).

H3. Bank concentration has a significant relationship with financial stability.

Besides the above-mentioned bank-specific variables, capital adequacy ratio (CRAR) and efficiency were also tested for their relationship with financial stability.

4. Data and methodology

4.1 Data

The study deploys several types of data and variables to gauge the link between NBFIs and financial stability.

4.1.1 Financial stability. The stability of the financial system is measured in terms of banking sector stability and overall stability. As is evident, banks are an important constituent of any financial system. Thus, the stability of the banking system reflects the soundness of the overall financial system. Banking stability is proxied using a z score similar to Diallo and Mansour (2017) and Uhde and Heimeshoff (2009). It is a consolidated score which measures the probability of default of a country's banking system and was sourced from Global Financial Development database (GFDD) of the World Bank.

In order to develop an overall measure of financial stability, an index measure was followed. The literature signifies the limitation of a single measure to proxy financial stability, and thus, FVI was constructed based on Aikman *et al.* (2017), Lee, Posenau, and Stebunovs (2017) and Lepers and Serrano (2020). The index consists of 12 indicators from four sectors: banking sector, non-financial sector, external sector and valuation pressures. The data for the indicators were sourced from the Bank of International Settlements (BIS), International Monetary Fund (IMF) and World Bank (WB). There was a paucity of data for the sample countries for some of the indicators employed in the literature, such as sovereign risk and housing pressures, and thus, only 12 indicators could be employed.

4.1.1.1 Financial vulnerability index. In order to build the FVI, the 12 indicators enumerated in Table 1 were standardized similar to Lepers and Serrano (2020). Therefore, principal component analysis was employed on these standardized indicators and weighted index [3] was calculated (eigenvalues are explained in Table 2). This index was used as the FVI and deployed as a measure of overall financial instability.

4.1.2 Growth of NBFIs. The growth of NBFIs is measured as growth of OFI assets and growth of the share of OFIs in TFA of a nation. It is alternatively narrowed down to the growth of ratio of second economic function (EF2). EF2 is a measure of FSB for analysing NBFIs, which includes entities like finance companies, consumer credit companies and leasing companies. We expect the growth of NBFIs to be negatively related to banking stability as found by Diallo and Mansour (2017) and positively related to FVI. The data for OFI assets and EF2 assets were sourced from the global monitoring report on NBFIs of the FSB.

4.1.3 Bank-specific variables. The study controls for bank-specific factors by deploying the following variables: net interest margin (NIM), regulatory capital ratio (CRAR), concentration and efficiency. The following data were sourced from the GFDD, World Bank.

Banks' NIM is a measure of banking sector profitability deployed in the literature (Ozili, 2018). Thus, the higher the NIM, the higher the banking stability.

Bank concentration is measured using two proxies similar to those of Uhde and Heimeshoff (2009). The relationship between bank concentration and financial stability can be positive or negative due to "concentration-stability" (positive) and "concentration-fragility" views (negative) in the literature.

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S. No.	Indicators	Components	Sign	Data source	References
1	*Banking sector*				
i	Non-performing loans (NPL)	Bank NPL to gross loans	Positive	Global Financial Development Database, World Bank (GFDD, WB)	Lepers and Serrano (2020)
ii	Capital	Bank capital to total assets	Negative	GFDD, WB	Lepers and Serrano (2020)
iii	Credit disbursed	Bank credit to GDP	Positive	GFDD, WB	Lee *et al.* (2017)
iv	Liquidity	Ratio of liquid assets to deposits and short-term funding	Negative	GFDD, WB	–
v	Interconnectedness	Consolidated foreign claims to GDP	Positive	Bank of International Settlements (BIS)	Lee *et al.* (2017)
2	*External sector*				
i	Current account balance	Current account balance as a percent of GDP	Negative	World Development Indicators, World Bank (WDI, WB)	Lee *et al.* (2017)
ii	Reserves	Total reserve to GDP	Negative	WDI, WB	Lee *et al.* (2017)
3	*Non-financial sector*				
i	Household credit	Total credit disbursed to household sector as a percent of GDP	Positive	BIS credit series	Lee *et al.* (2017)
ii	Non-financial corporations credit	Total credit to non-financial corporations as a percent of GDP	Positive	BIS credit series	Lee *et al.* (2017)
iii	National savings	Gross domestic savings as a percent of GDP	Negative	International Financial Statistics (IFS), IMF	Lepers and Serrano (2020)
4	*Valuation pressures*				
i	Stock market capitalization	Ratio of market capitalization of listed domestic companies to GDP	Positive	WDI, WB	Lepers and Serrano (2020)
ii	Stock price volatility	Average of the 360-day volatility of the national stock market index	Positive	GFDD, WB	Lepers and Serrano (2020)

Source(s): Authors' own creation

Table 1.
Indicators used to construct financial vulnerability index

Banks are required to maintain capital against the risks assumed by them which is determined by the CRAR. A higher CRAR safeguards banks against these risks and thereby enhances banking and overall stability. Thus, banking stability and CRAR are expected to be positively related.

In order to measure efficiency, banks' overhead costs-to-total assets (TA) ratio and banks' cost-to-income ratio were deployed. The literature evidenced cost-to-income ratio and banking stability to be negatively related (Uhde & Heimeshoff, 2009). A lower cost-to-income ratio will improve banks' efficiency and thus increase banks' stability.

Institutional quality is measured using the World Governance database from the World Bank. A composite indicator was constructed using these components referred to as the

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Components	Eigenvalue	Proportion	Cumulative
1	3.03998	0.2533	0.2533
2	1.68476	0.1404	0.3937
3	1.36893	0.1141	0.5078
4	1.11501	0.0929	0.6007
5	0.925897	0.0772	0.6779
6	0.89566	0.0746	0.7525
7	0.775134	0.0646	0.8171
8	0.640314	0.0534	0.8705
9	0.512374	0.0427	0.9132
10	0.402495	0.0335	0.9467
11	0.378057	0.0315	0.9782
12	0.261385	0.0218	1.0000

Source(s): Authors' own creation

Table 2. Eigenvalues of principal component analysis for the financial vulnerability index

World Governance Index (Gupta & Kashiramka, 2020). The literature has evidenced the role of institutional factors on banking stability (Demirgüç-Kunt & Detragiache, 1998; Ozili, 2018). They posit that the former can regulate the risk-taking behaviour of banks and thus improve banking stability. Thus, a positive relationship is expected between institutional quality and financial stability.

Several macro-economic variables such as bond market development, inflation and economic growth were controlled for. Table 3 lists the variables deployed in the study.

4.2 Econometric specification

In order to gauge the relationship between the growth of NBFIs and financial sector stability, the model as deployed by Diallo and Mansour (2017) is proposed as follows:

$$FS_{it} = \gamma_1 + \gamma_2 NBFI_{it} + \sum_{k=1}^{K} \gamma_k X_{kit} + \theta_i + \lambda_t + \varepsilon_{it} \quad (1)$$

where FS_{it} is referred to as the financial stability for country i during the year t. It is measured using the logarithm of the Z-score of return on assets of the banking sector for country i during the year t and is a measure of financial sector stability similar to Diallo and Mansour's (2017) and Uhde and Heimeshoff's (2009) measure of financial stability. Financial stability is also measured using the FVI, which is constructed using 12 indicators based on the methods followed by Aikman *et al.* (2017), Lee *et al.* (2017) and Lepers & Serrano (2020).

$NBFI_{it}$ measures the growth of NBFI assets for country i during the year t. X_{kit} is a set of k control variables. These control variables include bank-specific variables, macroeconomic variables and institutional quality. θ_i *and* λ_t are the country-fixed effects and time-fixed effects, and ε_{it} is the error term.

4.3 Methodology

The present study employs panel data since it enables an examination of different cross sections over time. Moreover, it takes into account unobserved heterogeneity. Panel data are generally estimated using fixed-effect or random-effect models. The fixed-effect model is estimated in such a way that it cancels out the effect of unobserved heterogeneity in the model, while the random-effect model assumes that unobserved heterogeneity is uncorrelated with the explanatory variables.

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S.No.	Variables	Explanation	Source
A	*Dependent variable*		
Financial stability			
i	Bank Z score	It is the probability of default of a country's banking system. It is calculated as $\frac{ROA+\frac{Equity}{Assets}}{sd(ROA)}$ sd(ROA)[a] is the standard deviation of ROA	Global Financial Development Database, World Bank (GFDD, WB)
ii	Financial vulnerability index (FVI)	Principle component analysis of 12 indicators from four sectors, namely banking, external sector, non-financial sector and stock markets. It captures the vulnerability in these sectors and is a measure of financial instability	Authors calculation
B	*Explanatory variables*		
i	NBFI		
a	YOYOFI/TFA	Log of year-on-year growth rate of the proportion of financial assets of other financial institutions (YOYOFI) in total financial assets of a nation (TFA)	Global Monitoring Report on NBFI 2021, FSB
b	YOYNBFI lending	YOY growth rate of lending to the NBFI sector	BIS
c	YOYEF2/TEF	Log of year on year growth rate of the proportion of financial assets of EF2 (YOYEF2) to financial assets of total economic function (TEF) .	Global Monitoring Report on NBFI 2021, FSB
C	*Control variables*		
i	Bank concentration		
a	Asset concentration(3) (Bank3Conc)	Assets of the three largest banks as a proportion of total assets of the commercial banking system	GFDD, WB
b	Asset concentration(5) (Bank5Conc)	Assets of the five largest banks as a proportion of total assets of the commercial banking system	GFDD, WB
ii	Bank efficiency		
a	Cost-to-income ratio (BankC_I)	It is measured as the ratio of operating expenses of a bank to the sum of net-interest revenue and other operating income	GFDD, WB
b	Overhead costs to total assets (%) (BankO_HCost/TA)	Ratio of operating expenses of a bank to the value of all assets held	GFDD, WB
iii	Bank profitability		
	Net interest margin (BankNIM) (%)	Percentage of bank's net interest revenue as a percentage of its average interest-bearing (total earning) assets	GFDD, WB
iv	Bank regulatory ratio		
	Capital adequacy ratio (BankCRAR)	The percentage of total regulatory capital to total assets (risk-weighted)	GFDD, WB
v	Liquidity		
	Liquid assets to deposits and short-term funding (%)	It is the ratio of liquid assets to deposits and short-term funding. Liquid assets include cash and bank balances, trading securities, loans and advances to banks and cash collaterals. Deposits and short-term funding includes total customer deposits and short-term borrowing	WDI, WB

Table 3.
List of variables deployed in the study

(*continued*)

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S.No.	Variables	Explanation	Source
D	*Country-level variables*		
i	Economic growth (GDP)	Annual percentage change in real gross domestic product	IFS, IMF
ii	Stock market development (StockTr)	Yoy growth of stocks traded divided by their respective market capitalization	WDI, WB
iii	Bond market development (OSpublicDebt)	Ratio of outstanding public debt securities to GDP. It covers long-term bonds and notes, treasury bills, commercial paper and other short-term notes	GFDD, WB
iv	Policyrate	The rate used by Central Bank of a nation to determine the monetary policy	IFS, IMF
v	Inflation	The growth rate (yoy) of GDP implicit deflator[b]	Data Mapper, IMF
vi	Institutional quality World Governance Index (WGI)	Principle component analysis of indicators of governance namely; regulatory quality, government effectiveness, political stability, voice and accountability, rule of law and control of corruption	World Governance Indicators (WGI), WB
vii	Crisis	Dummy for the GFC of 2008. It is equal to 1 for the period 2007–2009	

Note(s):
[a]The ROA, equity and assets are aggregated on a country level using unconsolidated data of banks from Bankscope and Orbis database. The sd (ROA) is calculated for country-years with at least 5 bank-level observations
[b]The implicit deflator was calculated as the ratio of GDP in current local currency to GDP in constant local currency (base year: 2010)
Source(s): Authors' own creation

Table 3.

The extant study investigates NBFIs in 11 EMEs for the period 2002–2020. Thus, we have a case of macropanels (Baltagi, 2008; Cameron & Trivedi, 2009). In such cases, there may be non-stationarity as a result of the large number of time series for each cross section.

Furthermore, macropanels may suffer from cross-section dependence (CD) in the data. Thus, the study deploys CD tests and panel unit root tests.

The issue of non-stationarity makes it vital to apply panel unit root tests on the variables. The extant literature has developed numerous panel unit root tests. The application of appropriate test depends on whether the data are balanced or unbalanced panels. Also, there are tests designed to check panel unit root in the presence of CD.

Thus, the present study applies the Phillips–Perron test, Fisher-type augmented Dickey–Fuller test, Maddala and Wu (1999) and Im–Pesaran–Shin (IPS) test (as we have unbalanced panels). Moreover, the study also deploys Pesaran (2007) as it accounts for the assumption of CD.

The null hypothesis in the case of all the above tests assumes variables to be non-stationary.

In order to test for autocorrelation, the Wooldridge test was applied. The null hypothesis of this test is based on the assumption of errors to be serially uncorrelated (Wooldridge, 2012). The modified Wald test was deployed to test for heteroscedasticity.

The errors were found to be auto-correlated and heteroskedastic. Also, the data had CD. Thus, the models were estimated using the panel-corrected standard error (PCSE) method.

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Due to the limited number of time series observations per cross section, dynamic methods such as panel auto-regressive distributed lag model could not be applied.

5. Results

Table 4 reports the descriptive statistics of the panel data deployed in the study. All the variables were winsorized at 95%. Variables such as bank concentration and GDP displayed non-stationarity (Table A1 in the appendix displays results of panel unit root tests). The non-stationary variables were stationary at the first order, and the required differencing was done for these variables. Table A2 in the appendix displays CD results based on Pesaran (2015), and Table A3 presents the correlation among the explanatory variables. Thus, the PCSE estimation method was deployed to control for heteroscedasticity, CD and serial correlation.

Tables 5 and 6 enumerate the estimation results.

Variables	N	Mean	StdDev	Minimum	Median	Maximum
BankZscore	197	2.488	0.516	1.401	2.621	3.143
FVI	174	0.00084	0.432	−0.846	−0.070	1.535
YOYOFI/TFA	195	5.468	11.860	−15.003	2.877	29.076
YOYEF2/TEF	139	−2.009	10.885	−23.056	−1.815	20.026
YOYOFI	195	25.396	27.202	−3.386	16.857	111.595
YOYEF2	139	15.904	15.531	−6.605	13.796	52.022
Bank3Conc	194	52.365	17.316	25.615	52.001	80.444
Bank5Conc	192	67.571	19.300	35.137	68.124	99.544
BankO_Hcost/TA	191	3.330	1.884	1.105	2.967	8.099
BankC_I	194	51.853	10.417	33.694	52.328	69.066
Bank NIM	191	4.340	1.618	2.343	3.841	8.203
BankROA	191	1.846	0.922	0.542	1.671	4.118
BankNPL/grossLoans	187	3.574	2.304	1.132	2.857	9.461
StockTr	185	62.960	59.470	6.856	34.413	205.017
Liquidity	188	31.702	20.080	9.844	22.311	80.562
GDP	198	3.952	3.716	−4.107	4.398	9.997
Inflation	198	7.603	5.129	−0.003	6.417	18.014
OSpublicDebt	160	23.812	17.095	2.566	18.002	62.899
BankCRAR	184	15.952	2.782	12.189	15.884	21.904
Policyrate	193	6.923	4.771	0.250	6.250	18.000
WGI	198	0.000	1.000	−1.511	−0.189	2.758
Crisis	198	0.167	0.374	0.000	0.000	1.000

Note(s): The descriptive statistics, namely, mean, standard deviation (StdDev), minimum, median and maximum of the variables deployed in the study are reported. They include log of bank Z score, FVI called as the Financial Vulnerability Index, year on year (yoy) OFI financial assets to TFA (YOYOFI/TFA), YOYEF2/TEF, which is the growth rate of ratio of financial assets of EF2 to financial assets of total economic function (TEF), yoy growth of OFI financial assets (YOYOFI), yoy growth of financial assets of EF2 (YOYEF2), Bank3conc refers to assets of 3 largest banks, Bank5conc refers to assets of 5 largest banks, BankO_Hcost/TA refers to ratio of bank overhead cost to total assets, BankC_I refers to cost to income ratio, BankNIM refers to NIM ratio of banks, BankROA refers to return on assets, BankNPL/grossLoans refers to the ratio of non-performing loans (NPL) to gross loans, stock turnover ratio (StockTr), the ratio of liquid assets to deposits and short-term funding (liquidity), GDP implies yoy growth of real GDP, the yoy growth rate of inflation (Inflation), outstanding public debt to GDP (OSpublicDebt), BankCRAR refers to capital adequacy ratio of banks, Central Bank policy rate, WGI refers to the world governance index and Crisis represents the dummy for global financial crisis of 2008.

Source(s): Authors' own creation

Table 4.
Descriptive statistics

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Variables Dependent variable	Expected sign	(1)	(2)	(3)	(4)	(5)	(6)	(7)
					Bank Zscore			
YOYOFI/TFA	+/−	−0.0006**	−0.00057**	−.00077**	−0.00014	0.00002		
		(0.0003)	(0.0003)	(0.0003)	(0.00061)	(0.00043)		
YOYEF2	+/−						−0.00094	
							(0.00076)	
YOYEF2/TEF	+/−							0.00022
								(0.0006)
Bank3Conc	+/−	−0.00033			−.00026	−0.00049	−.00051	−0.00033
		(0.00059)			(0.0014)	(0.00069)	(0.0009)	(0.0008)
Bank5Conc	+/−		−0.0005	−0.00017				
			(0.0009)	(0.0009)				
BankC_I	−		−0.0057***	−0.0053***	−0.0042**	−0.0046***	−0.001	0.00097
			(0.0018)	(0.002)	(0.002)	(0.0017)	(0.0023)	(0.002)
Bank O_H cost/TA	−	−0.012*						
		(0.007)						
BankNIM	+	0.047***	0.031***	0.03***	0.026***	0.035***	0.048***	0.048***
		(0.006)	(0.007)	(0.011)	(0.0085)	(0.0077)	(0.011)	(0.099)
BankCRAR	+	0.019***	0.019***	0.027***	0.019***	0.02***		0.025***
		(0.0053)	(0.0049)	(0.0061)	(0.0058)	(0.0047)		(0.0062)
GDP	+	−0.0009	−9.0e-06	−0.0014	−.00035	0.00052	−0.0025	−0.0013
		(0.0015)	(0.0016)	(0.0021)	(0.0017)	(0.0014)	(0.003)	(0.002)
Inflation	−	−0.00023	0.00010	−0.00047	0.00011	−0.00009	−0.0066**	−0.0018
		(0.0015)	(0.0013)	(0.0017)	(0.0015)	(0.0012)	(0.0026)	(0.0018)
StockTr	+			0.0004				
				(0.0003)				
Liquidity	+				0.0009			
					(0.0015)			
OSpublicDebt	−			0.0011				
				(0.0016)				
WGI	+					0.073***		
						(0.017)		
Crisis	−	−0.046*	−0.038	−0.028	−0.029	−0.033	−0.032	−0.049**
		(0.028)	(0.025)	(0.03)	(0.029)	(0.021)	(0.029)	(0.024)
Constant		1.40***	1.80***	1.70***	1.70***	1.70***	1.90***	1.30***
		(0.09)	(0.15)	(0.18)	(0.17)	(0.13)	(0.18)	(0.18)
Number of observations		164	165	134	162	167	136	125
Number of countries (N)		11	11	11	11	11	11	11
Country Effects		Yes	Yes	Yes	Yes	Yes	Yes	Yes
R squared		0.95	0.95	0.95	0.94	0.95	0.96	0.97
Wald Chi-Square		18701***	19268***	7749***	8253***	14815***	9414***	410084***

Note(s): ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. The models are estimated using PCSE to account for heteroscedasticity, cross-section dependence and serial correlation

The results for association between shadow banking growth and banking stability are presented using Log of Bank Z score (BankZscore) as the dependent variable. The explanatory variables include the NBFI growth measured as the growth rate of the ratio of OFI financial assets to total financial assets (YOYOFI/TFA), yoy growth of financial assets of EF2 (YOYEF2) and growth rate of ratio of financial assets of EF2 to financial assets of total economic function (YOYEF2/TEF). Bank-specific variables include bank concentration of three largest banks (Bank3Conc), bank concentration of five largest banks (Bank5Conc), bank cost-to-income ratio (BankC_I), bank overhead cost to total assets (BankO_Hcost/TA), BankNIM and capital adequacy ratio (BankCRAR). Macro-economic variables include GDP, Inflation, StockTr, Liquidity, ratio of outstanding public debt to GDP (OSpublicDebt), World Governance Index (WGI) and crisis dummy

Source(s): Authors' own creation

Table 5.
Association between NBFI growth and bank stability

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Variables Dependent variable	Expected sign	(1)	(2)	(3)	(4)	(5)	(6)	(7)
					FVI			
YOYOFI/TFA	+/−	0.0031** (0.0013)	0.003** (0.0014)	0.0026** (0.0012)	0.0026** (0.0012)	0.0039** (0.0019)		
YOYEF2	+/−						0.0028** (0.0001)	
YOYEF2/TEF	+/−							0.0024* (0.0014)
Policyrate			0.0018 (0.0042)				0.00029 (0.0044)	
Bank3Conc	+/−			0.00017 (0.0012	0.00029 (0.0013)	0.00068 (0.0017)		0.0019 (0.002)
BankC_I	+			0.0022 (0.0029)	0.002 (0.0029)	−0.0033 (0.016)		0.0042 (0.0043)
BankCRAR	−			−0.019* (0.01)	−0.018* (0.01)	0.002 (0.015)		
BankNIM	−			−0.029** (0.013	−0.029** (0.013)	−0.033** (0.016)		
GDP	−	0.000018 (0.0043)	0.00024 (0.0043)	−0.0045 (0.0038)	−0.0042 (0.0039)	−0.00041 (0.0051)	0.0012 (0.0037)	−0.0018 (0.007)
Inflation	+	−0.00048 (0.0047	−0.00095 (0.0038)	−0.0032 (0.0033)	−0.0029 (0.0034)	−0.0043 (0.0037)	−0.0042 (0.0048)	
WGI	−				0.045 (0.04)			
Crisis	−					−0.023 (0.10)		
Constant		−0.31*** (0.11)	−0.38*** (0.11)	−0.21 (0.23)	−0.2 (0.23)	0.11 (0.37)	−0.5*** (0.12)	−0.57***
Number of observations		168	165	152	152	152	138	112
Number of countries		11	11	11	11	11	11	11
Country effects		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time effects		Yes	Yes	Yes	Yes	No	Yes	Yes
R squared		0.54	0.56	0.57	0.58	0.39	0.61	0.55
Wald Chi-Square		346935***	411032***	140253***	152298***	279***	236572***	3.9e+07***

Note(s): ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. The models are estimated using PCSE estimation

The results for association betwwen NBFI growth and overall stability using the financial vulnerability index (FVI) as a dependent variable is presented. The explanatory variables include the shadow banking growth measured as the growth rate of ratio of OFI financial assets to total financial assets (YOYOFI/TFA), yoy growth of financial assets of EF2 (YOYEF2) and growth rate of the ratio of financial assets of EF2 to financial assets of total EF (YOYEF2/TEF). Bank-specific variables include bank concentration of three largest banks (Bank3Conc), bank cost-to-income ratio (BankC_I), BankNIM and bank capital adequacy ratio (BankCRAR). Macro-economic variables include GDP, Inflation, central bank policy rate, World Governance Indexr (WGI) and crisis dummy

Source(s): Authors' own creation

Table 6.
Association between NBFI growth and overall stability

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The growth of NBFIs (measured as the growth rate of the ratio of OFI financial assets to TFA) has a significant negative association with the bank Z score. This finding supports the first hypothesis of the study. This implies that the higher the growth of NBFIs, the lower the banking stability, highlighting the risks associated with NBFIs. The findings are consistent with the results of Diallo and Mansour (2017), who found a negative effect of shadow banking on financial stability. However, no significant results are obtained in the case of finance company's assets (EF2). Furthermore, financial instability measured using the FVI shows a positive association with the growth of NBFIs. Also, the EF2 assets are positively and significantly related to financial instability. Thus, it can be said that the growth of shadow banks can be detrimental to the financial stability of a nation.

Furthermore, we also controlled for bank-specific and institutional variables for measuring the link between the shadow banking sector and financial stability. Bank profitability measured in terms of the NIM ratio has the expected positive sign, and thus, the higher the profitability of banks, the higher their stability. Our results are aligned with those of Ozili (2018). This finding confirms our second hypothesis. In the case of the FVI, the NIM ratio is significantly negatively related to financial instability. Thus, the lower the banks' profitability, the higher the instability in the financial system. Bank concentration has a negative association with bank stability, albeit insignificant. In the case of the FVI, it is positive but insignificant. This result implies that higher concentration leads to financial fragility, but no conclusive evidence could be drawn for the relationship between bank concentration and financial stability. Thus, our third hypothesis is not validated. Our results support the findings of Ijtsma *et al.* (2017). Banks' regulatory ratio, i.e. CRAR, positively and significantly relates to the bank Z score. This finding implies that the higher the amount of capital maintained by banks, the higher the banking stability. The results are in line with the findings of Uhde and Heimeshoff (2009). The capital adequacy of banks is negatively and significantly related to the FVI, thus having similar implications. Bank efficiency (measured in terms of bank cost-to-income ratio and bank overhead cost-to-total assets ratio) is negatively related to bank Z score. This implies that the lower the cost-to-income ratios of banks, that is, the higher the banking efficiency, the higher is the banking stability. The results for the bank cost-to-income ratio are insignificant in the case of the FVI index. Institutional quality (proxied as the World Governance Indicator index) is positively related to banking stability. Thus, the higher the government effectiveness and other institutional indicators of a country, the higher the stability of the banking system. This result supports the findings of Ozili (2018). However, no significant results are found in the case of the FVI. The crisis dummy for the GFC 2008 negatively relates to banking stability. This result highlights the lower stability caused by the crisis. Furthermore, we also measured the association between the central bank policy rate and financial vulnerability, but no conclusive evidence could be drawn for the same.

6. Robustness analysis

In order to validate the findings of the study, several robustness checks were deployed.

Firstly, we deploy fixed-effect regression where standard errors are accounted for using Driscoll & Kraay (1998) estimation. It allows for serial correlation, CD and heteroscedasticity in the data. Similar signs and results were obtained for the explanatory variables as obtained using the PCSE method (Tables 7 and 8).

Secondly, the findings of the study are consistent across both the measures of financial stability, i.e. Bank Z score and the constructed FVI. The results are consistent across both the estimation methods as well.

Thirdly, the NBFI is alternately measured as growth in lending provided to NBFIs. Table 9 summarizes the results of the relationship between the growth of lending to shadow

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Variables	Expected sign	(1)	(2)	(3)	(4)	(5)	(6)	(7)
Dependent variable		Bank Zscore						
YOYOFI/TFA	+/−	−0.001**	−0.00096***	−0.0012**	−0.00061	−0.00041		
		(0.0004)	(0.0003)	(0.0004)	(0.00038)	(0.00038)		
YOYEF2	+/−						−0.0011	
							(0.0009)	
YOYEF2/TEF	+/−							0.00039
								(0.0009)
Bank3Conc	+/−	−0.0003			−0.00009	−0.0002	−0.001	−0.00038
		(0.0008)			(0.00077)	(0.0008)	(0.0009)	(0.0009)
Bank5Conc	+/−		−0.00044	0.0001				
			(0.001)	(0.001)				
BankO_Hcost/TA	−	−0.0093						0.00096
		(0.0093)						(0.0017)
BankC_I	−		−0.0061**	−0.0061*	−0.0063**	−0.0048*	−0.00078	
			(0.0026)	(0.0029)	(0.0027)	(0.0024)	(0.0024)	
BankNIM	+	0.053***	0.038***	0.039***	0.039***	0.045***	0.054***	0.052***
		(0.005)	(0.008)	(0.010)	(0.006)	(0.0077)	(0.01)	(0.005)
BankCRAR	+	0.021***	0.02**	0.029***	0.021***	0.021**		0.018***
		(0.0058)	(0.0074)	(0.007)	(0.0074)	(0.0073)		(0.0038)
GDP	+	−0.0014	−0.00027	−0.0024	0.00024	0.00025	0.00011	−0.0011
		(0.0015)	(0.0013)	(0.0016)	(0.0018)	(0.0013)	(0.0016)	(0.0007)
Inflation	−	−0.0002	0.00011	−0.0004	0.00022	2.5e-06	−0.0012	−0.00058
		(0.0007)	(0.00067)	(0.0012)	(0.00083)	(0.0005)	(0.0015)	(0.0009)
StockTr	+			0.0005				
				(0.0004)				
Liquidity	+				0.0023**			
					(0.0008)			
OS publicDebt	−			0.0005				
				(0.0011)				
WGI	+					0.075**		
						(0.019)		
Crisis	−	−0.059**	−0.052*	−0.043	−0.049*	−0.044**		
		(0.027)	(0.025)	(0.024)	(0.025)	(0.02)		
Year							0.012	0.012***
							(0.002)	(0.002)
Constant		2***	2.3***	2.2***	2.3***	2.2***	−21***	−22***
		(0.091)	(0.14)	(0.10)	(0.13)	(0.1)	(4)	(4.3)
Number of observations		164	165	134	162	168	136	125
Number of countries		11	11	10	11	11	11	11
Country effects		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time effects		No	No	No	No	No	Yes	Yes
Within *R* squared		0.29	0.31	0.28	0.29	0.3595	0.41	0.444
F		74.39***	26.01***	17.55***	44***	28***	24***	36***

Note(s): ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. The models are estimated using the fixed-effect method, and standard errors are corrected using the Driscoll–Kraay method. It accounts for heteroscedasticity, cross-section dependence and serial correlation

The results for association between shadow banking growth and banking stability is presented using Log of Bank Z score (BankZscore) as the dependent variable. The explanatory variables include the NBFI growth measured as the growth rate of the ratio of OFI financial assets to total financial assets (YOYOFI/TFA), yoy growth of financial assets of EF2 (YOYEF2) and growth rate of the ratio of financial assets of EF2 to financial assets of total economic function (YOYEF2/TEF). Bank-specific variables include bank concentration of three largest banks (Bank3Conc), bank concentration of five largest banks (Bank5Conc), bank cost-to-income ratio (BankC_I), bank overhead cost to total assets (Banko_hcost/TA), bank net interest margin (BankNIM) and bank capital adequacy ratio (BankCRAR). Macro-economic variables include GDP, Inflation, StockTr, Liquidity, ratio of outstanding public debt to GDP (OSpublicDebt), World Governance Index (WGI) and crisis dummy

Source(s): Authors' own creation

Table 7.
Robustness tests for association between NBFI growth and banking stability (estimation using Driscoll–Kray method)

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Variables Dependent variable	Expected sign	(1)	(2)	(3)	(4)	(5)	(6)	(7)
					FVI			
YOYOFI/TFA	+/−	0.0067*** (0.0016)	0.0053** (0.0017)	0.0031 (0.0019)	0.0066*** (0.0017)	0.0067*** (0.0017)		
YOYEF2	+/−						0.0025* (0.0012)	
YOYEF2/TEF	+/−							0.0031** (0.0014)
Policyrate			−0.0071 (0.008)				−0.015 (0.009)	
Bank3Conc	+/−			0.0018 (0.0011)	0.00081 (0.00083)	0.00083 (0.0008)		0.0005 (0.0012)
BankC_I	+			−0.0041 (0.0042	−0.0011 (0.0046)	−0.00071 (0.0041)		0.0051 (0.0048)
BankCRAR	−				−0.0002 (0.018)	−0.00095 (0.017)		−0.0005 (0.019)
BankNIM	−			−0.043** (0.015)	−0.018 (0.017)	−0.017 (0.018)		0.031 (0.018)
GDP	−	−0.0037 (0.0031)	−0.0025 (0.0024)	−0.0009 (0.005)	−0.0021 (0.0026)	−0.0021 (0.0027)	0.0022 (0.0023)	0.0037 (0.0026)
Inflation	+	0.0007 (0.0029)	−0.0003 (0.0024)	0.00002 (0.0038)	0.00083 (0.003)	0.00082 (0.003)	−0.0015 (0.0028)	−0.0039 (0.0038)
WGI	−					0.024 (0.053)		
Crisis	+			−0.087 (0.075)			−0.20 (0.079)	
Year		0.035*** (0.0056)	0.038*** (0.0053)		0.034*** (0.006)	0.033*** (0.0051)	0.039 (0.0052)	0.037*** (0.0059)
Constant		−70*** (11)	−76*** (11)	0.28 (0.21)	−68*** (11)	−67*** (10)	−0.8** (0.31)	−75*** (12)
Number of observations		168	165	130	152	152	124	124
Number of countries		11	11	10	11	11	11	11
Country effects		Yes	Yes	Yes	Yes	Yes	Yes	Yes
Time effects		Yes	Yes	No	Yes	Yes	No	Yes
Within R squared		0.28	0.32	0.1285	0.26	0.26	0.31	0.30
F		18.43***	20.23***	13.26***	23***	45***	60.62***	33.24***

Note(s): ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. The models are estimated using the fixed-effect method, and standard errors are corrected using the Driscoll–Kraay method The results for association between shadow banking growth and overall stability are presented using the Financial Vulnerability Index (FVI) as the dependent variable. The explanatory variables include the shadow banking growth measured as the growth rate of ratio of OFI financial assets to total financial assets (YOYOFI/TFA), yoy growth of financial assets of EF2 (YOYEF2) and growth rate of the ratio of financial assets of EF2 to financial assets of total EF (YOYEF2/TEF). Bank-specific variables include bank concentration of three largest banks (Bank3Conc), bank cost-to-income ratio (BankC_I), bank net interest margin (BankNIM) and bank capital adequacy ratio (BankCRAR). Macro-economic variables include GDP, Inflation, central bank policy rate, World Governance Index (WGI) and crisis dummy.

Source(s): Authors' own creation

Table 8. Robustness tests for association between NBFI growth and overall stability (estimation using the Driscoll–Kray method)

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banks and stability. We found that the growth of lending to NBFIs is negatively and significantly associated with banking stability (controlling for country and time effects). However, no conclusive evidence could be drawn for the relationship between lending to NBFIs and financial vulnerability.

7. Conclusion

NBFIs are important financial institutions in catering to the needs of an economy. However, their unchecked growth and growing interdependence with OFIs might cause the financial system to suffer. In light of this, the present study analyses the association between the growth of the NBFI sector and financial stability for 11 EMEs for the period 2002–2020. The study further narrows down its scope to finance companies since they are an important type of NBFIs in EMEs. In order to measure the overall stability, an FVI representing the instability of four sectors is constructed.

The study finds that the growth of the shadow banking sector negatively relates to banking stability and positively relates to the FVI. This implies that the higher the growth of the shadow banks, the higher the financial fragility. As Bengtsson (2013) pointed out the

Variables	(1)	(2)	(3)	(4)
Dependent variable	BankZscore		FVI	
YOY NBFI Lending	−0.0009*	−0.001**	−0.00012	−0.000087
	(0.00048)	(0.00045)	(0.000092)	(0.000092)
Bank3Conc	−0.0015*	−0.0009	0.00044	−0.00015
	(0.0009)	(0.0007)	(0.0012)	(0.0014)
BankC_I	−0.0043***	−0.0033**	0.003	0.0013
	(0.0017)	(0.0016)	(0.0029)	(0.0029)
GDP	0.0009	0.0014	−0.0049	−0.0018
	(0.002)	(0.0018)	(0.0039)	(0.0043)
Inflation	−0.0017	0.00002	−0.0037	−0.0023
	(0.0012)	(0.0012)	(0.0035)	(0.0039)
BankCRAR		0.019***	−0.021	
		(0.0046)	(0.011)	
BankNIM	0.037***	0.035***	−0.025	−0.027
	(0.0065)	(0.0066)	(0.013)	(0.013)
Constant	1.9***	1.6***	0.086	−0.12
	(0.13)	(0.13)	(0.28)	(0.25)
Number of observations	181	169	152	163
Number of countries	11	11	11	11
Country effects	Yes	Yes	Yes	Yes
Time effects	Yes	Yes	Yes	Yes
R squared	0.94	0.96	0.56	0.52
Wald Chi-Square	1102087***	3279894***	127652***	174027***

Note(s): ***, ** and * indicate significance at the 1%, 5% and 10% levels, respectively. The models are estimated using PCSE estimation

The results for association between shadow banking growth and financial stability are presented. The dependent variable, financial stability, is measured using log of Bank Z score (banking stability) in models 1-4 and Financial Vulnerability Index (FVI -overall stability) in models 5-8. The explanatory variables include the shadow banking growth measured as year-on-year lending to NBFIs as a ratio to GDP. Bank-specific variables include bank concentration of three largest banks (Bank3conc), bank cost-to-income ratio (BankC_I), bank net interest margin (BankNIM) and bank capital adequacy ratio (BankCRAR). Macro-economic variables include GDP and Inflation

Source(s): Authors' own creation

Table 9. Robustness tests using growth of lending to NBFIs as the dependent variable

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price instability in case of MMFs, Bouguelli (2020) discussed the financial instability of SPVs and Huang (2018) highlighted the interconnectedness and pro-cyclicality of shadow banks. Also, the study confirms the theoretical findings of Barth *et al.* (2015) who discussed that higher shadow banking creates financial fragility in the financial system. NBFIs are pro-cyclical in nature, that is, they assume innumerable risks in the boom periods and suffer in the bust periods (Chaturvedi & Singh, 2022). Furthermore, the study finds a positive relationship between the growth of EF2 assets and overall instability. The EF2 assets majorly represent finance companies. Thus, the growth of finance companies increases the financial vulnerability of EMEs. The findings are aligned with Chaturvedi and Singh (2022), who report NBFCs to be interconnected and posing a threat of systemic risk (in the Indian context). Bank-specific variables such as cost-to-income ratio, regulatory ratio and NIM indicate similar relationships as in prior literature. Institutional quality is positively associated with the bank stability, highlighting the role of governance in ensuring the stability of banks.

The study thus contributes to the novel literature on shadow banking, especially in EMEs. NBFIs are being regulated and monitored, but there are several risks that these entities are prone to, which may further inhibit the financial stability of economies. Thus, their unchecked growth and interlinkage with other entities in the financial system need to be monitored and regulated. The FVI can serve as a tool to measure the likely risks of financial instability. The scope of the study is limited to 12 indicators for the development of the FVI due to the non-availability of data for the sampled EMEs. Furthermore, the likely impact of regulations imposed by the nations on NBFIs can also be examined.

Notes

1. However, the emerging economies, namely Argentina, Indonesia, Saudi Arabia and Turkey, constituted lesser share than 20%.
2. The three countries were sourcing more than 50% of their fund requirements from banks in 2019 (FSB, 2020). In their recent report, the FSB has not reported the composition of liabilities of finance companies.
3. First five components were retained based on the Kaiser criteria as their eigenvalues were more than or equal to 1. Weighted index was constructed using the first five components. The weights used were the ratio of proportion explained by the component to total percentage explained by the five components.

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Appendix

Variables	IPS W t-bar	Fisher Phillips and Perron Inverse chi-square	Fisher ADF regression Inverse chi-square	Maddala and Wu (1999) Chi square	Pesaran (2007) Z t-bar
BankZscore	−1.40*	39.11**	35.95**	35.95**	−1.186**
YOYOFI/TFA	−8.14***	172.86***	128.32***	185.71***	−5.82***
YOYEF2/TEF	−8.56***	93.47***	112.27***		
GDP	1.48	25.21	16.12	16.12	1.12
Inflation	−0.90	64.43***	31.20*	31.20*	0.37
Bank3Conc	1.12	22.19	12.71	12.71	1.12
Bank5Conc	0.96	20.09	16.11	16.12	0.42
Bank ROA	−0.39	44.94***	28.67	29.08	0.63
Bank NIM	−0.33	47.49***	21.36	23.48	−1.75**
Bank C_I	−2.06**	61.45***	42.18***	40.93***	−2.06***
Bank O_H cost/ TA	−0.51	77.51***	27.01	27.01	1.10
Bank NPL/gross loans	−3.37***	214.27***	61.77***	57.80***	−1.39*
StockTr	−0.48	44.37***	23.12	21.25	2.45
OS publicDebt	0.48	61.35***	45.53***	6.40	−0.10
Liquidity	−0.53	19.0	24.02	34.23**	0.02
BankCRAR	−1.13	29.23	33.05*	33.05*	−0.23
Policyrate	−0.68	21.99	24.15	21.99	0.31

Source(s): Authors' own creation

Table A1. Results of panel unit root tests

Variables	CD-test	*p* value	corr	abs(corr)
FVI	12.8	0	0.555	0.621
BankZscore	7.45	0	0.334	0.459
YOYOFI	8.99	0	0.291	0.345
YOYOFI/TFA	3.95	0	0.162	0.337
YOYEF2	1.45	0.146	0.059	0.317
YOYEF2/TEF	2.44	0.015	0.113	0.327
Bank3Conc	−0.35	0.728	−0.013	0.444
Bank5Conc	−0.3	0.762	−0.01	0.415
BankROA	4.66	0	0.161	0.396
BankNIM	3.33	0.001	0.114	0.39
BankO_Hcost/TA	5.64	0	0.191	0.42
BankC_I	1.86	0.063	0.065	0.286
StockTr	2.87	0.004	0.087	0.334
BankNPL/GrossLoans	4.96	0	0.2	0.464
Liquidity	−0.56	0.578	−0.016	0.433
GDP	13.46	0	0.509	0.547
Inflation	6.4	0	0.239	0.365
OSpublicDebt	6.25	0	0.254	0.562
BankCRAR	3.17	0.002	0.133	0.368
Policyrate	5.29	0	0.207	0.438
Crisis	25.11	0	1	1

Source(s): Authors' own creation

Table A2. Results of cross-section dependence (CD) tests

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Table A3.
Correlation matrix

Variables	BanZ score	YOYOFI/TFA	Bank5Conc	BankC_I	BankNIM	BankCRAR	StockTr	OSpublicDebt	GDP	Inflation	Crisis
BankZscore	1										
YOYOFI/TFA	−0.0279	1									
Bank5Conc	0.3429	−0.0065	1								
BankC_I	−0.1978	−0.0388	−0.0554	1							
BankNIM	−0.4974	0.0832	−0.083	0.312	1						
BankCRAR	−0.3758	−0.009	0.0149	0.0974	0.4277	1					
StockTr	0.3384	0.0943	0.1167	−0.5473	−0.349	−0.1674	1				
OSpublicDebt	0.3395	−0.0491	0.3667	0.339	0.1495	0.1119	−0.0305	1			
GDP	0.0602	0.2271	−0.1633	−0.3378	−0.2678	−0.2489	0.4019	−0.1944	1		
Inflation	−0.3801	−0.0016	0.0052	0.1631	0.595	0.0868	−0.2261	−0.1436	−0.1471	1	
Crisis	0.0045	−0.0481	0.0288	−0.0002	−0.0041	0.0227	0.0981	−0.0543	−0.1235	−0.0159	1

Source(s): Authors' own creation

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